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BHARATIYA ANTARIKSH HACKATHON 2025

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Team name: **ISROnauts.AI**

Team Leader Name : **Nakibul Islam**

Problem Statement : **Chase the Cloud: Leveraging Diffusion Models for Cloud Motion Prediction using INSAT-3DR/3DS Imagery.**

Team Members

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Team Leader:

Name: **Nakibul Islam**
College: **National Institute of Technology Silchar**

Team Member 1:

Name: **Rana Talukdar**
College: **VIT Bhopal University**

Team Member 2:

Name: **Mainak Das**
College: **National Institute of Technology Silchar**

Team Member 3:

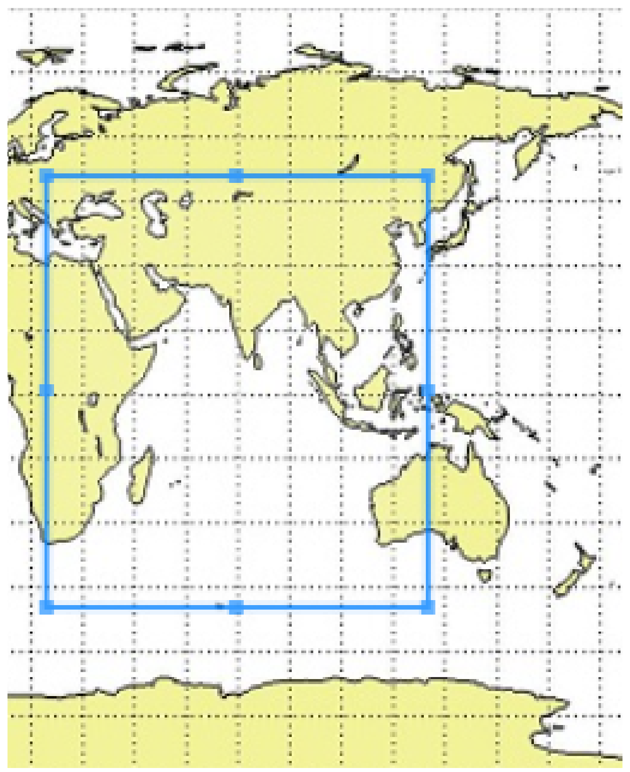
Name: **Kritiman Talukdar**
College: **National Institute of Technology Silchar**

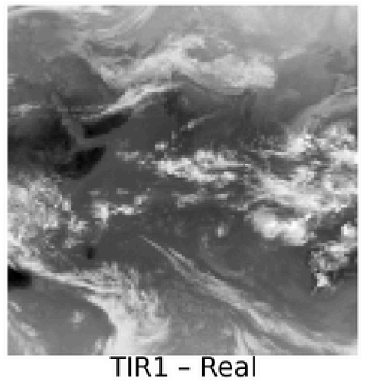
Brief about the Idea

Our project leverages conditional diffusion networks to forecast 0–1 hour cloud motion using multi-spectral INSAT-3DR/3DS imagery, delivering realistic future cloud states for improved nowcasting and early warning.

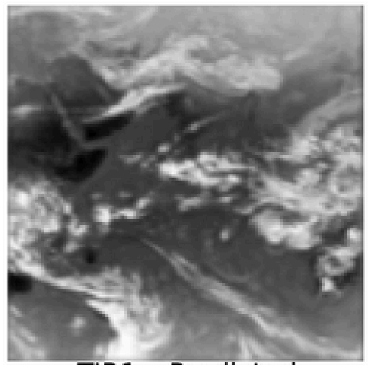
- A **novel 3D conditional diffusion model** with sinusoidal time-step embeddings captures complex spatio-temporal cloud dynamics
- **Multi-spectral input** (VIS, WV, TIR, SWIR) from indigenous INSAT satellites ensures rich contextual learning
- Outperforms traditional optical-flow and physics-based methods, achieving **high accuracy** in SSIM, MAE, and PSNR
- **Interactive web portal** enables users to select any location for bespoke short-term forecasts, compare predicted vs. real-time images with accuracy scores, and receive automated severe-weather alerts
- Ready for **seamless integration with meteorological agencies** to empower proactive disaster preparedness and response

INSAT-3DS L1C

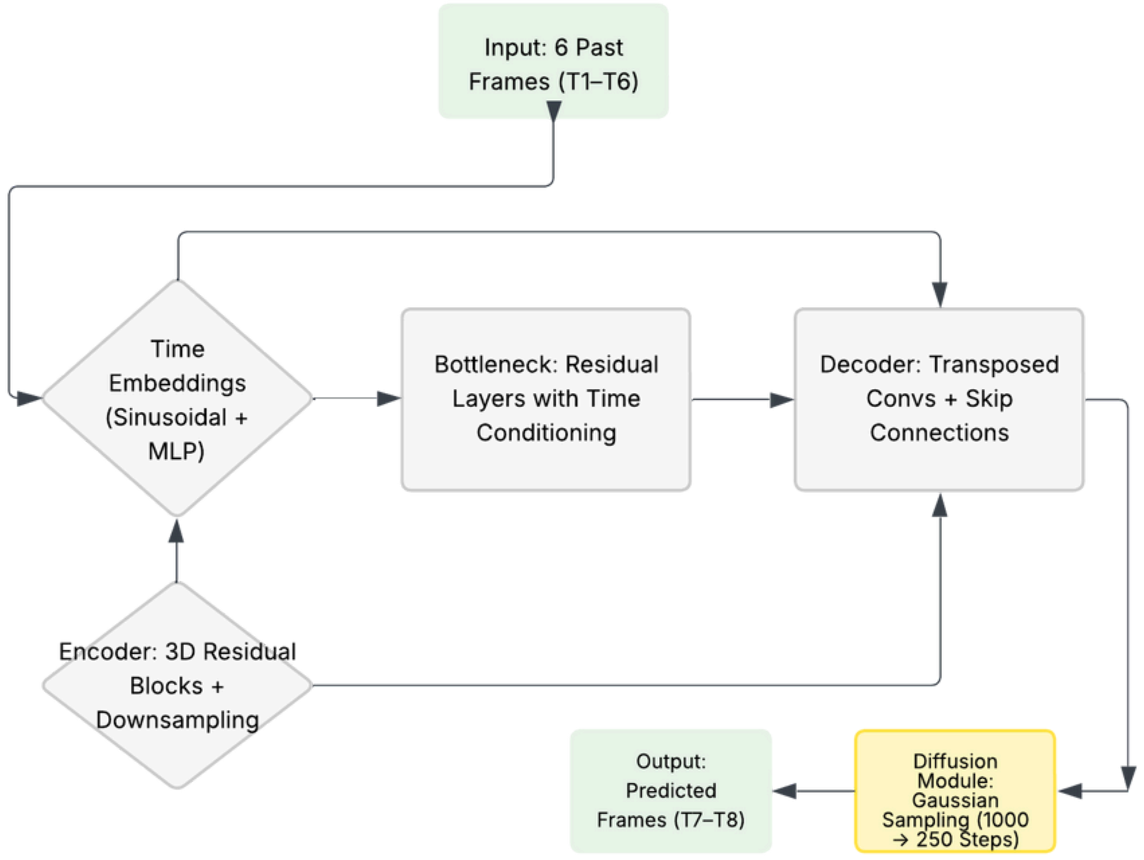




TIR1 – Real



TIR1 – Predicted



Opportunity & Differentiation

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How we differ

- Leverages a 3D U-Net to capture both spatial and temporal patterns in cloud motion
- Uses a Gaussian diffusion process to generate future satellite frames by denoising from noise
- Introduces learnable time-step embeddings at the encoder, bottleneck, and decoder stages
- Enables dynamic prediction adjustment throughout the diffusion process (time-aware conditioning)
- Time-aware conditioning improves temporal understanding; rarely used in traditional nowcasting
- Gradient Accumulation which facilitates less GPU computation

How it solves the problem

- Leverages past six multi-spectral INSAT frames in a 3D conditional diffusion model to produce realistic short-term (0–1 hr) cloud forecasts
- Uses indigenous INSAT imagery for region-specific accuracy over the Indian subcontinent
- Captures fine-scale cloud structures (storm tops, edges) and is validated using SSIM, MAE, and PSNR for reliable early warnings
- Generates multiple forecast outcomes to support scenario-based planning in aviation, solar energy, and disaster response
- Provides probabilistic outputs for risk-aware decision-making; useful for pilots, grid operators, and emergency responders

Our unique edge

- First-of-its-kind 3D diffusion model trained on indigenous INSAT imagery
- Integrated time-aware 3D UNet backbone tailored for spatio-temporal forecasting
- Web portal delivers real-time location-specific predictions and alert notifications
- Scalable for operational integration with IMD, NESAC, and disaster management agencies

List of Features Offered by the Solution

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- **Short-Term Forecasting**
  Predicts next 1–2 cloud frames from past 6 inputs (0–3 hr range), extendable to 4–6 frame predictions

- **Visual + Quantitative Evaluation**
  Animated predictions vs. ground truth + SSIM, MAE, PSNR metrics

- **Web-Based UI**
  Region selector + animated playback + live metrics display

- **Real-Time Storm Alerts**
  Detects IR cooling or WV spikes → in-app & email alerts

- **Scalable & Indigenous**
  Built on INSAT data, ready for IMD/NESAC integration

- **Historical Replay & Analytics**
  Archive past predictions vs. outcomes, show seasonal performance trends

- **Cross-Satellite Fusion**
  Fuse INSAT with polar-orbiter (FY-3/Copernicus) data

- **Chatbot Integration for Weather Interpretation**
  Enables users to query a built-in chatbot to understand predicted weather conditions (e.g., rainfall, storms) based on the model's output images.

- **Uncertainty Quantification**
  Show pixel-wise confidence maps (e.g. variance from diffusion samples) alongside each prediction

- **Ensemble Diffusion**
  Run multiple diffusion chains in parallel and average outputs to boost stability

- **Explainable Heatmaps**
  Generate Grad-CAM or attention maps to highlight cloud features driving each forecast

- **Real-Time Data Fusion**
  Ingest IMD ground-station sensors (wind, humidity) to refine alerts

- **Mobile Notifications & Dashboard**
  Push warnings via SMS/WhatsApp + responsive web view for on-the-go users

- **Open API & SDK**
  Expose a REST API + Python SDK so other teams/systems can integrate your forecasts

for higher-resolution snapshots

- **Collaborative Sharing**
  Allow users to annotate & share forecasts within the app (for research & training)

Use-case diagram

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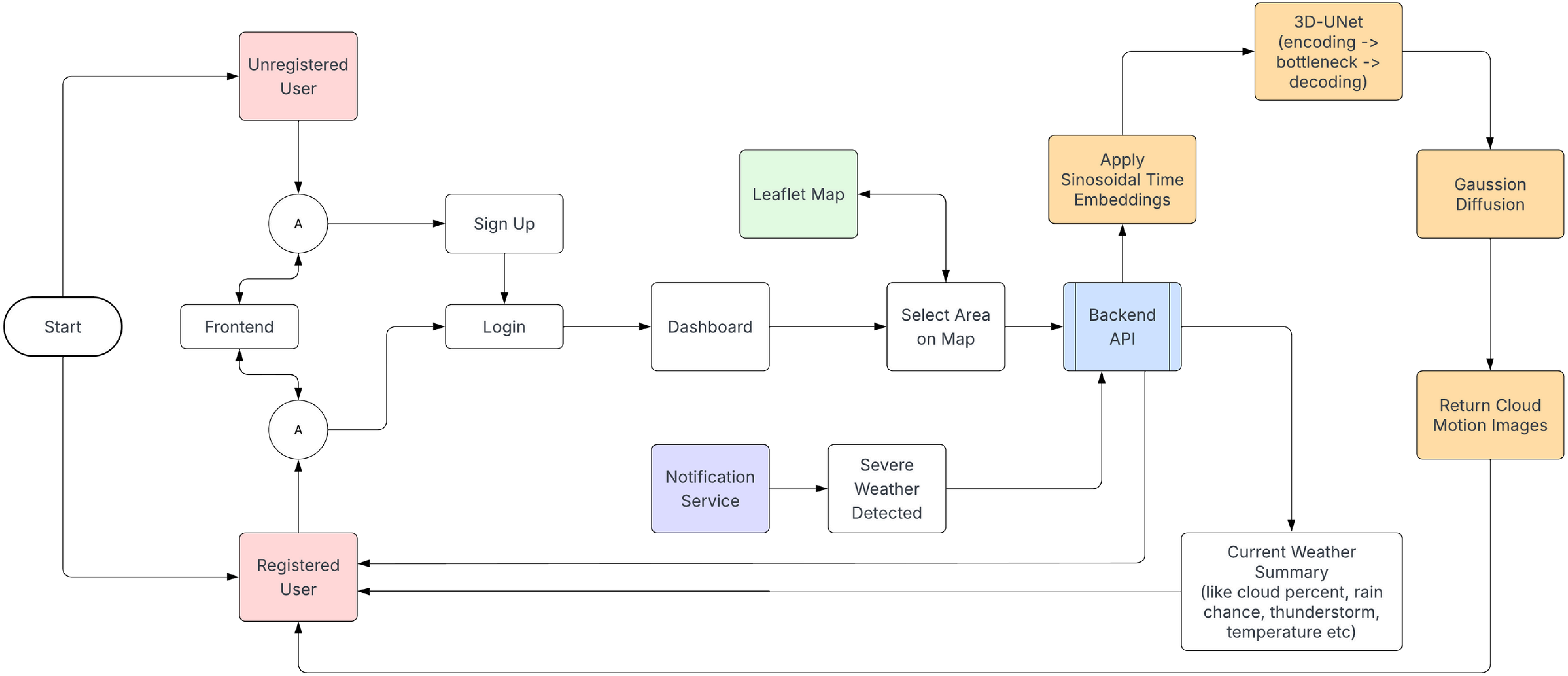
[Click here for GitHub repo]



Architecture diagram of the proposed solution

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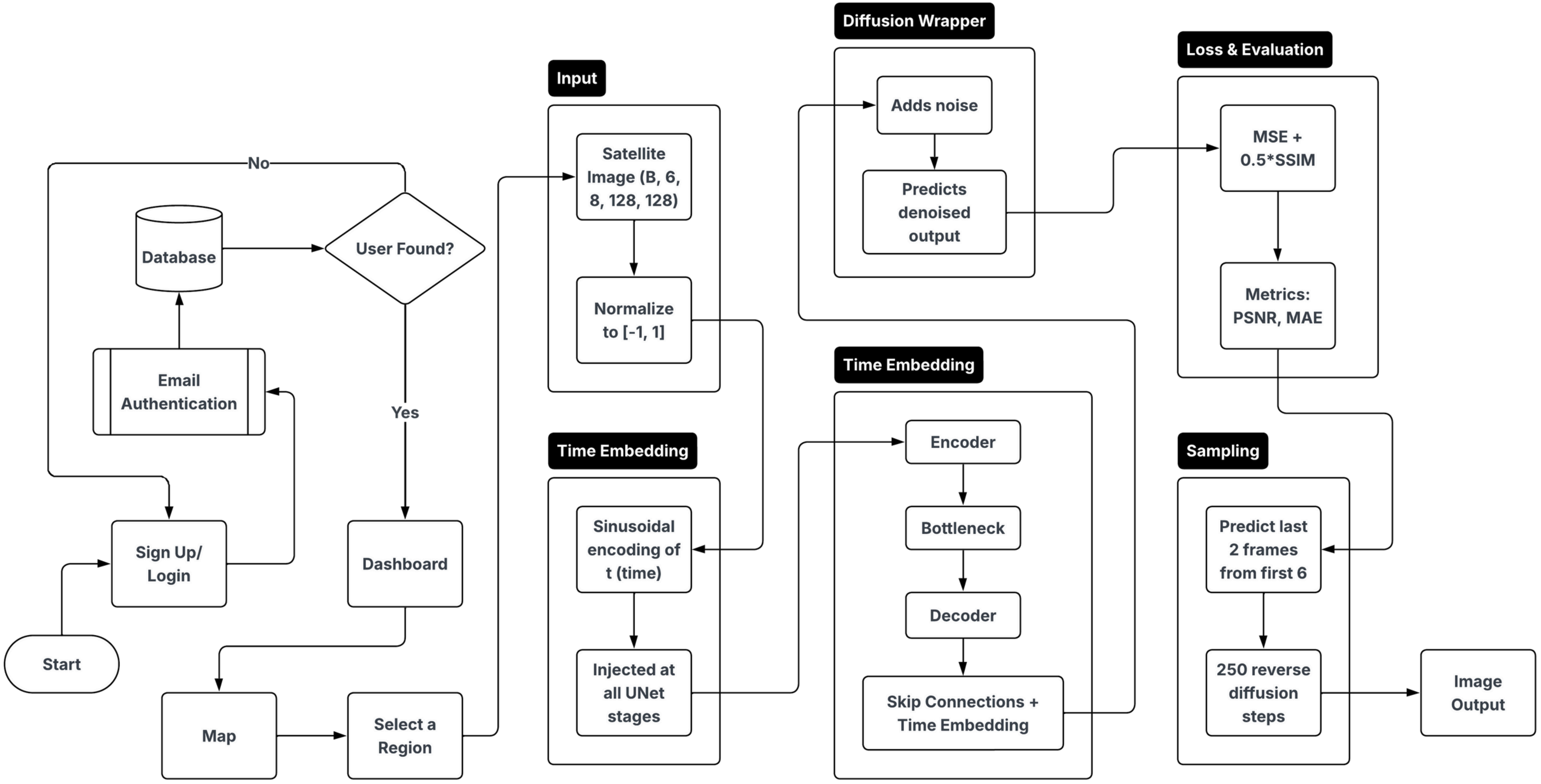

End-to-End Implementation Stack

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Tool Name	Use Case
PyTorch	Core deep learning framework for building & training the 3D conditional diffusion U-Net
Video-diffusion-pytorch	Plug-and-play PyTorch implementation of Gaussian diffusion for fast prototyping
TorchMetrics	Compute SSIM, MAE (and other quality metrics) in a differentiable, batched way
Docker	Containerize the model + server for consistent, scalable deployment with load-balancing
Matplotlib	Plotting and exporting “before vs after” comparison grids for slides and reports
Maps API	Interactive map UI for users to pick a location and fetch the corresponding satellite frames
Cloud Platform	Host inference service (Cloud Run / Compute Engine), storage (Cloud Storage), and autoscaling
Sveltekit + Tailwind + TypeScript + Vercel	Build a fast, responsive frontend for visualizing inputs, predictions, metrics and alerts

Estimated implementation cost

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Category	Details	Estimated Cost (INR)
1. Development & Engineering	1 AI + 1 web dev for 6 weeks (part-time, ₹600/hr avg)	₹72,000
2. Compute (Cloud/On-prem)	Colab Pro/GPU credits for 1 month + basic inference VPS	₹60,000
3. Storage & Data Transfer	1–2 TB hot storage + 3 TB transfer on free tier/S3-alike	₹15,000
4. Deployment & Monitoring	Docker + free-tier GCP/Vercel + open-source logging tools	₹10,000
5. Contingency & Misc (15%)	Buffer for GPU bursts, network spikes, small tools	₹23,000

Total Estimated Cost: ₹180,000 – ₹220,000

Predicted vs Actual Satellite Frames with Accuracy Metrics

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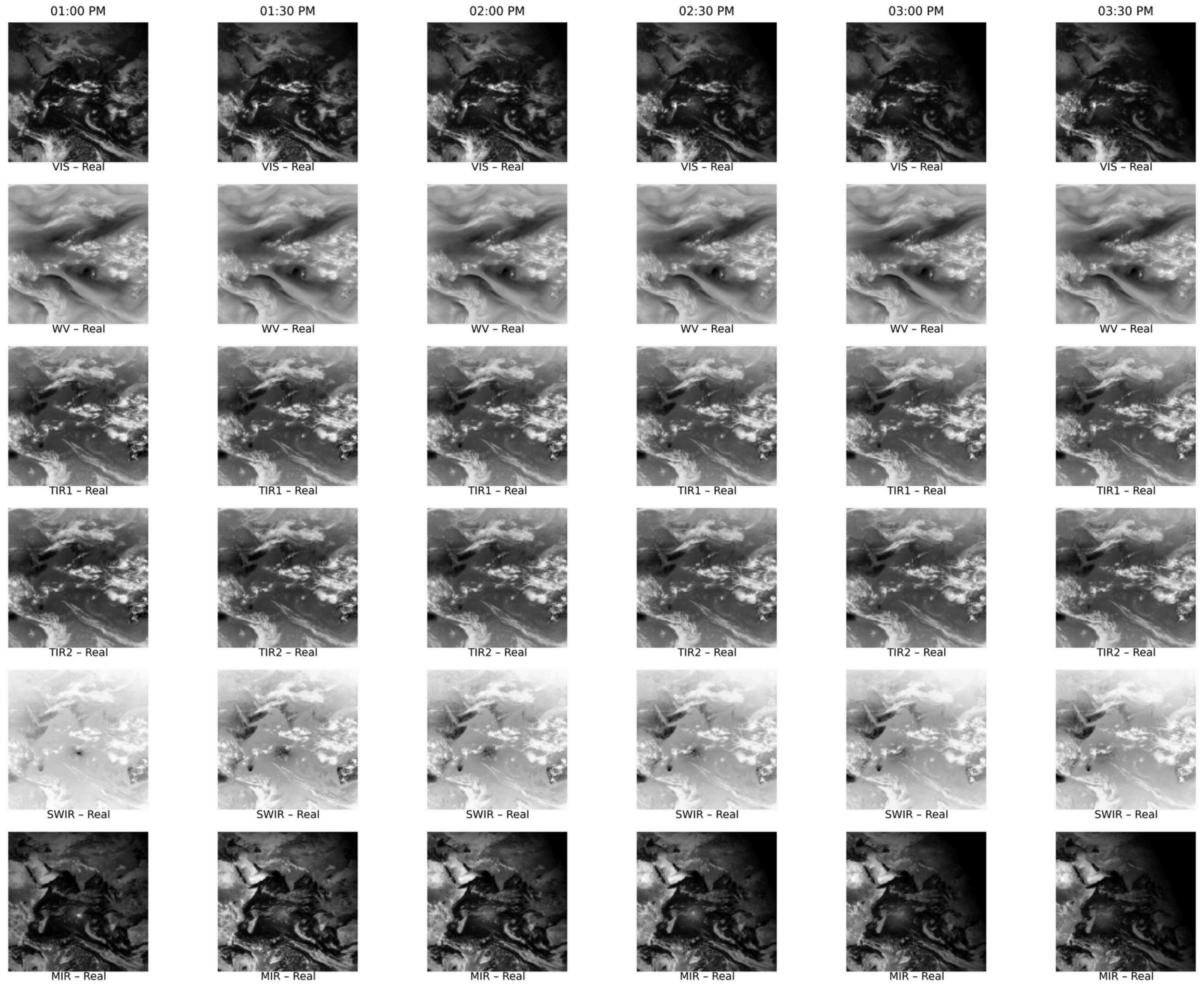
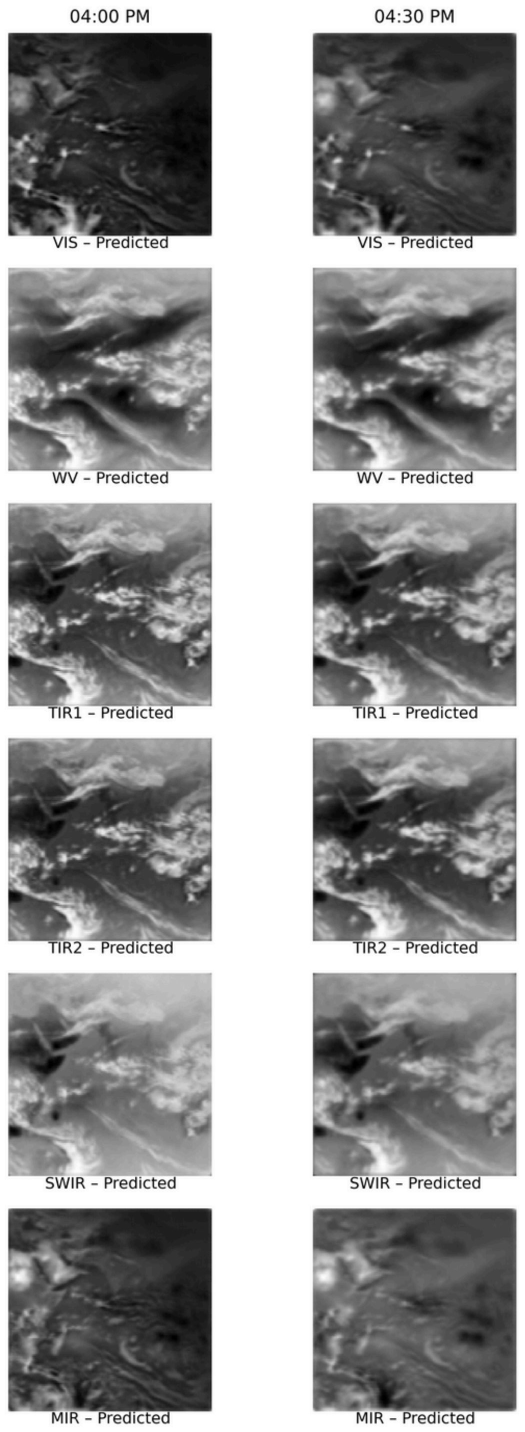
Actual frames



Predicted



Sequence Date: 01 Jan 2025

Table: T7 Accuracy Metrics

Band	SSIM	MAE	PSNR (dB)
VIS	0.6851	0.3362	14.34
WV	0.5834	0.1836	19.46
TIR1	0.6046	0.1427	20.81
TIR2	0.6115	0.1325	21.16
SWIR	0.8922	0.1232	23.14
MIR	0.6574	0.4123	12.89

Table: T8 Accuracy Metrics

Band	SSIM	MAE	PSNR (dB)
VIS	0.589	0.4049	12.52
WV	0.4124	0.2432	17.46
TIR1	0.4179	0.2283	17.42
TIR2	0.4614	0.1934	18.47
SWIR	0.8196	0.2156	18.62
MIR	0.5841	0.4559	11.86

Click here to check entire notebook

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THANK YOU